this year[4], and has inspired calculations by Weisskopf[5] and by Bethe[4].

According to Bethe's calculations, the shift of an ns level should become $W' \sim kZ^4/n^3$ cm.$^{-1}$, where the factor k is about 0·26 for $Z = 1$ and $n = 2$ and does not change very rapidly with Z and n. This agrees well with the observed level-shift.

This result can be applied to the ionization energy of helium-like atoms if as a first rough approximation the interaction between the two electrons is neglected. In the accompanying table I have calculated the ratio W'/Z^4, where W' is the difference between the calculated and the experimental ionization energy.

Z	W'	W'/Z^4
2	+ 14	0·88
3	+ 13	0·16
4	+ 84	0·33
5	+ 195	0·31
6	+ 311	0·24
7	+ 537	0·22
8	+ 1080	0·26
9	+ 1844	0·28
12	+ 4861	0·23
13	+ 4966	0·18

The value 0·88 obtained for $Z = 2$ is, of course, very uncertain, since the theoretical value lies within the experimental limits of error.

H. A. S. Eriksson

Institute of Mechanics and
Theoretical Physics,
Uppsala.
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Theory of the Anomalous Skin Effect in Metals

Pippard[1] has recently made an experimental and theoretical study of the skin effect in metals at high frequencies, in which he showed that the usual theory of the effect is no longer valid at low temperatures owing to the breakdown of Ohm's law when the free path of the conduction electrons becomes comparable with the penetration depth of the electric field. With the aim of putting Pippard's provisional theory of the anomalous effects on a more rigorous basis, we have reformulated his equations using the methods of the modern electron theory of metals, and we have succeeded in obtaining explicit solutions which make it possible to give a quantitative account of the phenomena to be expected over the whole frequency- and temperature-range.

We consider a semi-infinite metal and assume the electrons to be quasi-free, the energy being proportional to the square of the wave-vector. The relation between the current density and the (unknown) electric field at any point in the metal, for arbitrary values of the frequency ω and the free path l, can then be obtained by the usual statistical methods of the theory of metals based on the Boltzmann equation[2]. On combining this relation with Maxwell's equations, we obtain the following integro-differential equation for the electric field $f(x)$:

$$f''(x) = i\alpha \left\{ p \int_{-\infty}^{\infty} k(x-y)\, f(y)dy + (1-p)\int_0^{\infty} k(x-y)\, f(y)dy \right\}, \qquad (1)$$

where

$$k(u) = \int_1^{\infty} \left(\frac{1}{s} - \frac{1}{s^3}\right) \exp\{-s(1+i\omega\tau)|u|\}ds,$$

x is the distance into the metal measured in units of l, $\alpha = 3l^2/2\delta^2$ (δ being the classical skin-depth), τ is the time of relaxation of the electrons, and it has been assumed that a fraction p of the electrons is scattered specularly at the surface of the metal while the rest are scattered diffusely. Also, $f(x)$ is to be an even function of x. The basic equation of Pippard's treatment (*loc. cit.*, equation (7)) may be written in a form which differs only in minor detail from equation (1).

Equation (1) has been solved for the two limiting cases $p = 1$ and $p = 0$ by means of the theory of Fourier integrals[3]. The electric field is, in general, a very complicated function of x, but for comparison with experiment it is not necessary to know $f(x)$ explicitly since all observable quantities can be expressed in terms of the surface impedance Z, defined as $4\pi/c$ times the ratio of the electric to the magnetic field strength at the surface of the metal. For $p = 1$ we find that

$$Z = \frac{4\pi\omega l}{ic^2}\,\frac{f(0)}{f'(0)} = \frac{8i\omega l}{c^2(1+i\omega\tau)} \int_0^{\infty} \frac{dt}{t^2 + \dfrac{i\alpha}{(1+i\omega\tau)^3}\varkappa(t)}, \qquad (2)$$

where

$$\varkappa(t) = \frac{2}{t^3}\{(1+t^2)\tan^{-1} t - t\}.$$

The expression for $p = 0$ is more complicated, but the results differ only in detail from those obtained for $p = 1$.

For frequencies which are not too high, we have $\omega\tau \ll 1$, and the integral in (2) is then a function of the single parameter α. In this case equation (2) reduces to the classical expression $Z_{cl} = (\pi\omega l/c^2)\sqrt{(6/\alpha)}(1+i)$ when $\alpha \ll 1$, and to

$$Z_\infty = \frac{8\pi^{2/3}\omega l}{3\sqrt{3}\,.\,c^2\alpha^{1/3}}\,(1+\sqrt{3}\,i) \qquad (3)$$

when $\alpha \gg 1$. (For $p = 0$, Z_∞ has 9/8 times the value (3).) For intermediate values of α, Z has been obtained by numerical evaluation of the integral in (2). From the definition of α, it follows that α is proportional to l^3 for a given metal and a given frequency. Hence Z varies as $l^{-1/2}$ in the classical case but is independent of l when $\alpha \gg 1$. This result is in agreement with Pippard's observations of the resistive part of Z at a wave-length of 25 cm., and represents the quantitative refinement of his theoretical interpretation in terms of an 'ineffectiveness concept'[1]. From a detailed comparison of the experimental and theoretical curves, it is possible to estimate the value of α and hence of the free path corresponding to any particular temperature. Thus for Pippard's mercury specimen we find that $\alpha = 1{,}550$ and $l = 8·5 \times 10^{-4}$ cm. at $T = 3·72°$ K.

Writing $Z = R + iX$, we have $R = X$ in the classical limit, while equation (3) shows that

$R = X/\sqrt{3}$ when $\alpha \gg 1$. X has not been measured directly, but Pippard[4] has obtained some indirect experimental evidence which indicates that X does, in fact, exceed R in the anomalous region.

The expression (2) has further been computed as a function of $\omega\tau$ for several values of the ratio $\alpha/\omega\tau$. ($\alpha/\omega\tau$ is independent of ω and, for a given metal, determines the value of l; the values chosen correspond to several temperatures between room temperature and liquid helium temperatures.) The ratios R/R_{cl} and X/X_{cl} are found to have the classical value unity for sufficiently low, and again for sufficiently high, frequencies (which are, however, well below the frequency for which the metal becomes transparent), and to pass through a maximum value at intermediate frequencies. The anomalous effects become less marked as the temperature increases, and have disappeared almost completely at room temperature. To give some idea of the orders of magnitude involved, choosing values of the parameters which correspond to a very pure specimen of silver at liquid helium temperatures, we find that R/R_{cl} attains a maximum value of 180 at a wave-length of 69 μ, while X/X_{cl} attains a maximum value of 5·3 at a wave-length of 7·6 cm. The absorption coefficient of the metal is increased in the same ratio as R; thus in the above example the theoretical absorption coefficient at 69 μ is $1{\cdot}44 \times 10^{-3}$, whereas the classical value is only 8×10^{-6}. Since the absorption is so small even in the anomalous case, an experimental verification of these predictions is likely to be very difficult.

A full account of the theory is to be published elsewhere.

G. E. H. REUTER

Mathematics Department,
University of Manchester.

E. H. SONDHEIMER

Trinity College,
Cambridge.

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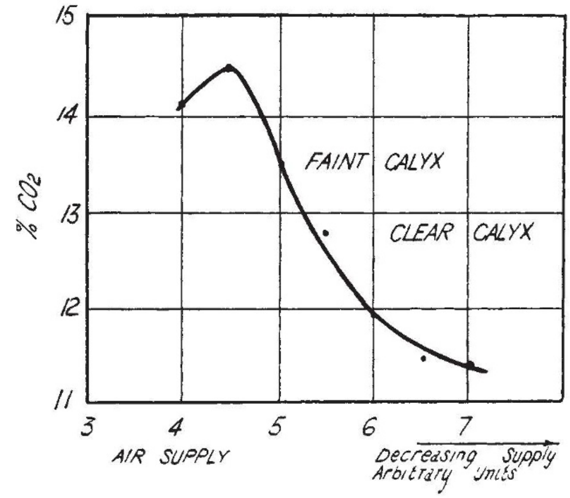
Use of 'Calyx' Burner to Determine Combustion Conditions

Brinsley and Stephens[1] have described an apparatus in which air and gas are pre-mixed and burned in a vessel from which the only exit is a small hole. When the proportion of air in a mixture originally air-rich is gradually reduced, the flame alters, and at one stage a violet sheath appears, surrounding the lower part of the flame. This sheath Brinsley and Stephens call a 'calyx'; they claim that it appears when the gas–air is that theoretically required for complete combustion.

It was thought that sampling the products of combustion as the theoretical mixture was burning would enable the maximum possible concentration of carbon dioxide in these products to be determined by a single analysis. This would be of service in testing gas appliances.

A mixture of coal gas and air was used. It was found that the flame is stable over only a small range of needle-valve settings: both the rate of supply of the mixture and its constitution have to be approximately correct or the flame goes out. Adjustment is not difficult once the operator has become familiar with the appearance of the flame in various circumstances, and the flame changes described by Brinsley and Stephens can then be readily observed.



Tests were made keeping the gas supply constant and cutting down the air in a series of arbitrary steps; at each setting a sample of the combustion products was taken and analysed for carbon dioxide. The results are shown in the accompanying graph. It will be seen that the calyx did not appear until after the maximum carbon dioxide concentration had been passed, and even then was only faintly visible. This reveals the chief objection to the use of the phenomenon for determination of theoretical combustion conditions. The calyx does not appear sufficiently suddenly; it 'fades in' as the air supply decreases, and the threshold visibility depends on the eyesight of the observer and on his experience. If it were used for determination of the maximum concentration of carbon dioxide in the combustion products, low results would be very probable.

H. ROSTRON HINDLEY

Physics Division,
Nash and Thompson, Ltd.,
Oakcroft Road,
Tolworth,
Surrey.

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A New Sulpha Compound ("6257") and its Use in Human Cholera Infection

It was observed by one of us (S. S. B.) in 1939 that hexa-methylene-tetra-amine, in a 10 per cent solution in normal saline, killed cholera vibrio in less than half an hour up to a concentration of 10^{12}. Although interrupted by conditions of field service, efforts resulted in the production of three crude compounds of hexamine linkage to *para*-aminobenzene sulphonamide, experiments with which, on animal and human cholera infection, gave hope of successful chemotherapy against Asiatic cholera. A chance discussion with the scientific department of C.I.B.A. (Basle) resulted in our obtaining a condensation product—"6257"—of two molecules of 2 p-amino-benzene sulphonamido-thiazole ('Cibazol') and three molecules of formaldehyde with the formula $C_{21}H_{22}O_6N_6S_4$ (the actual constitution not having yet been worked out) and molar weight much greater